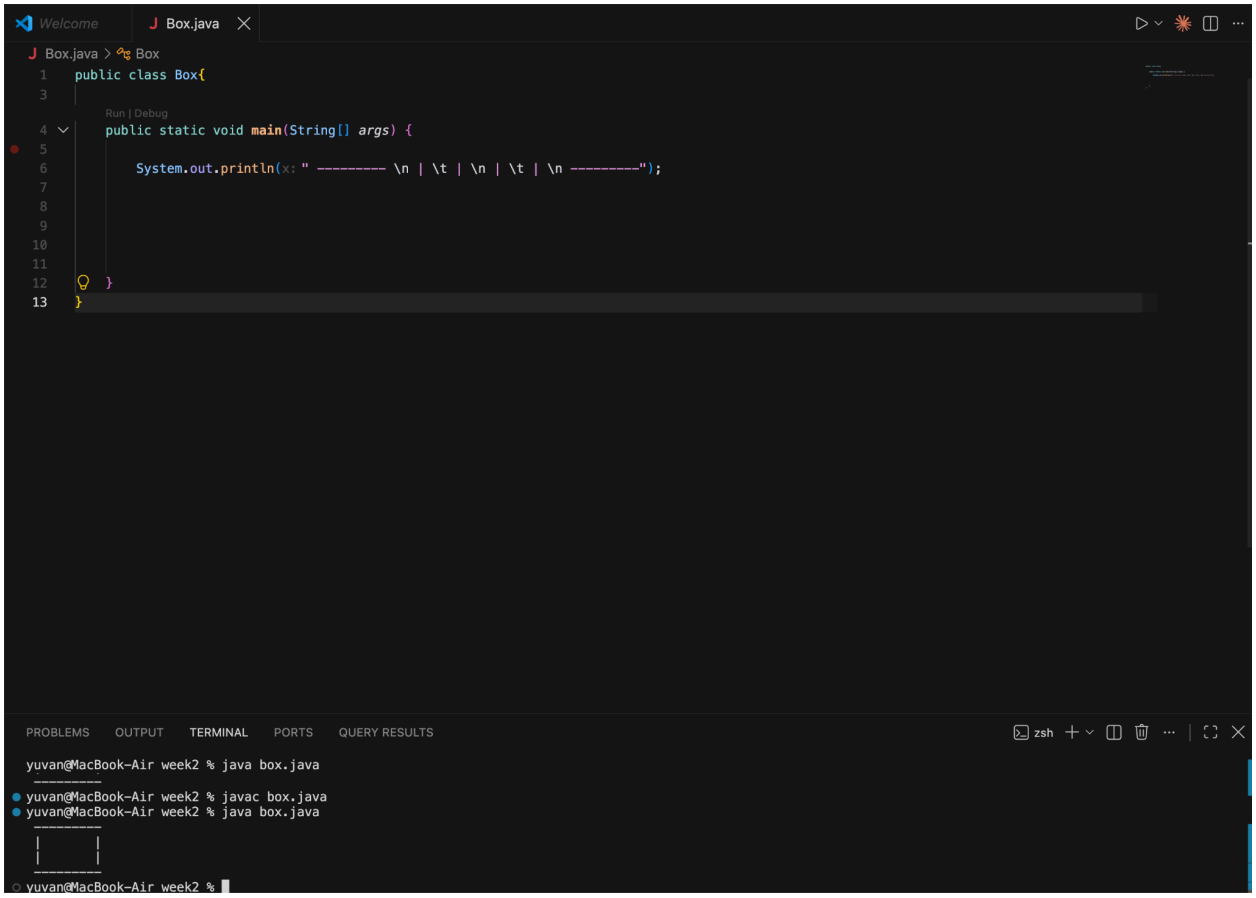
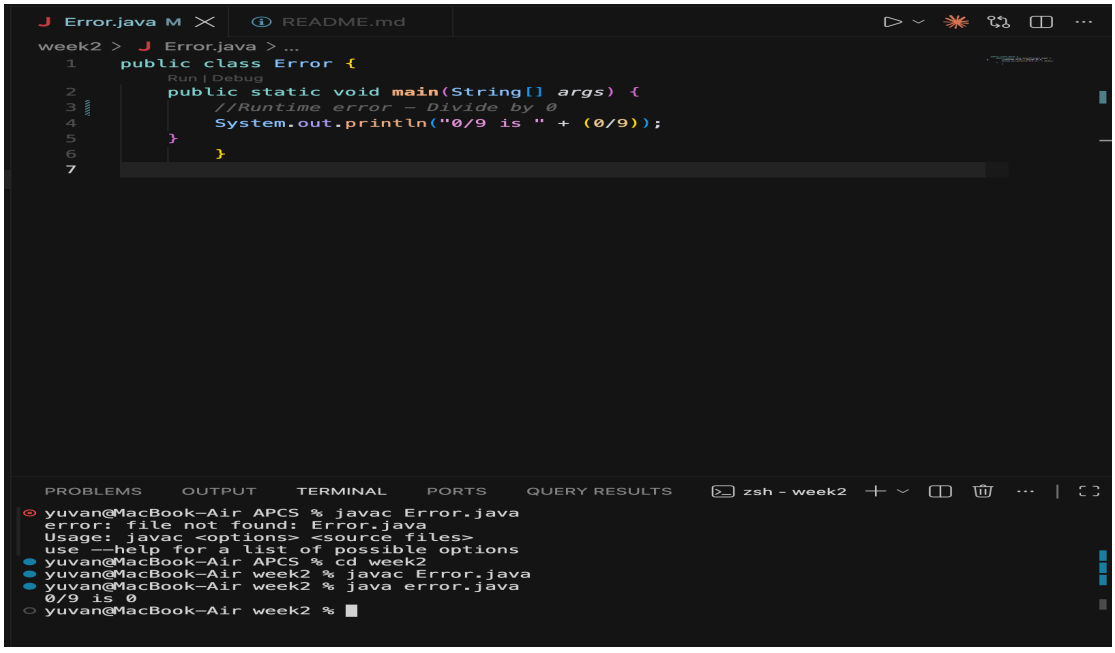
Warmup 8/17/26



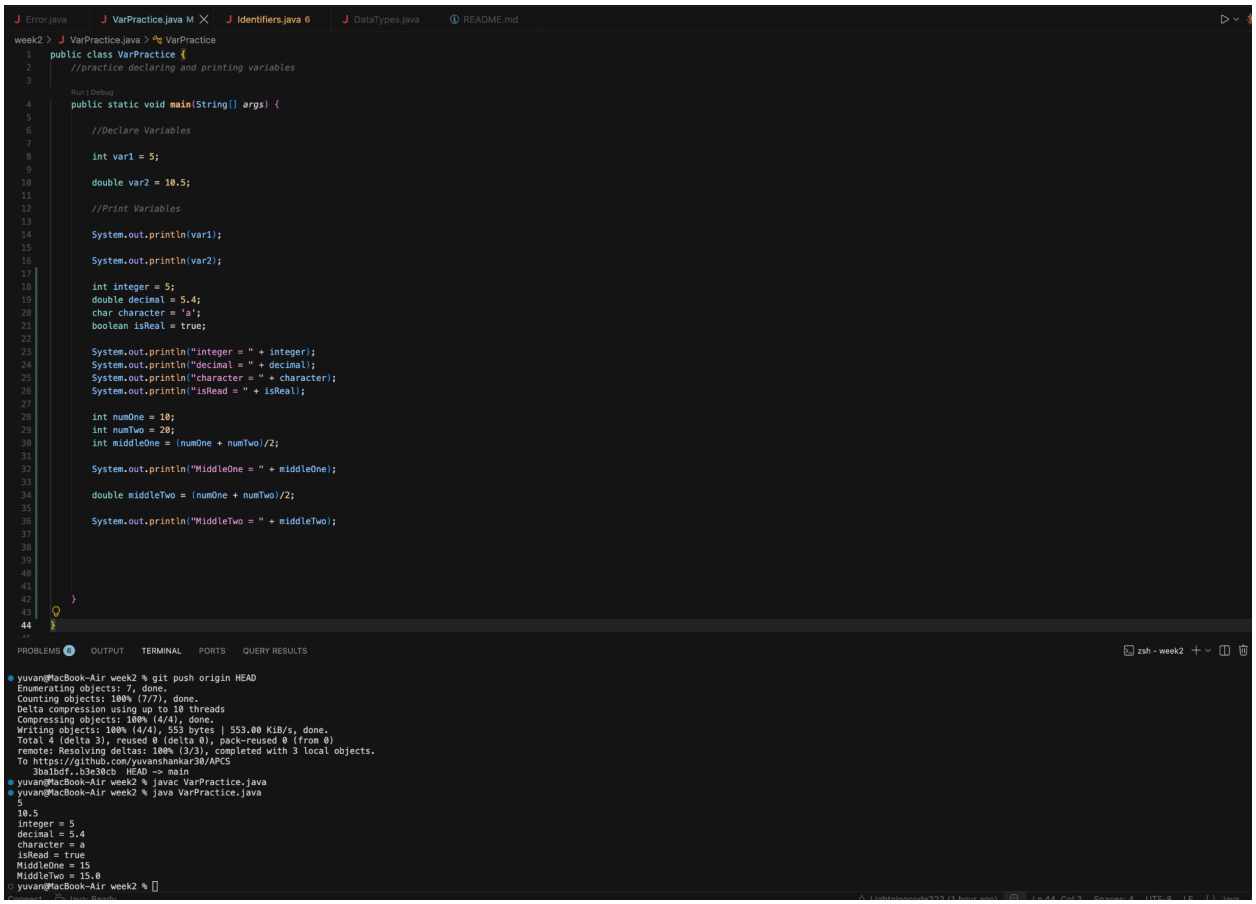
8/17/26 Classwork 1



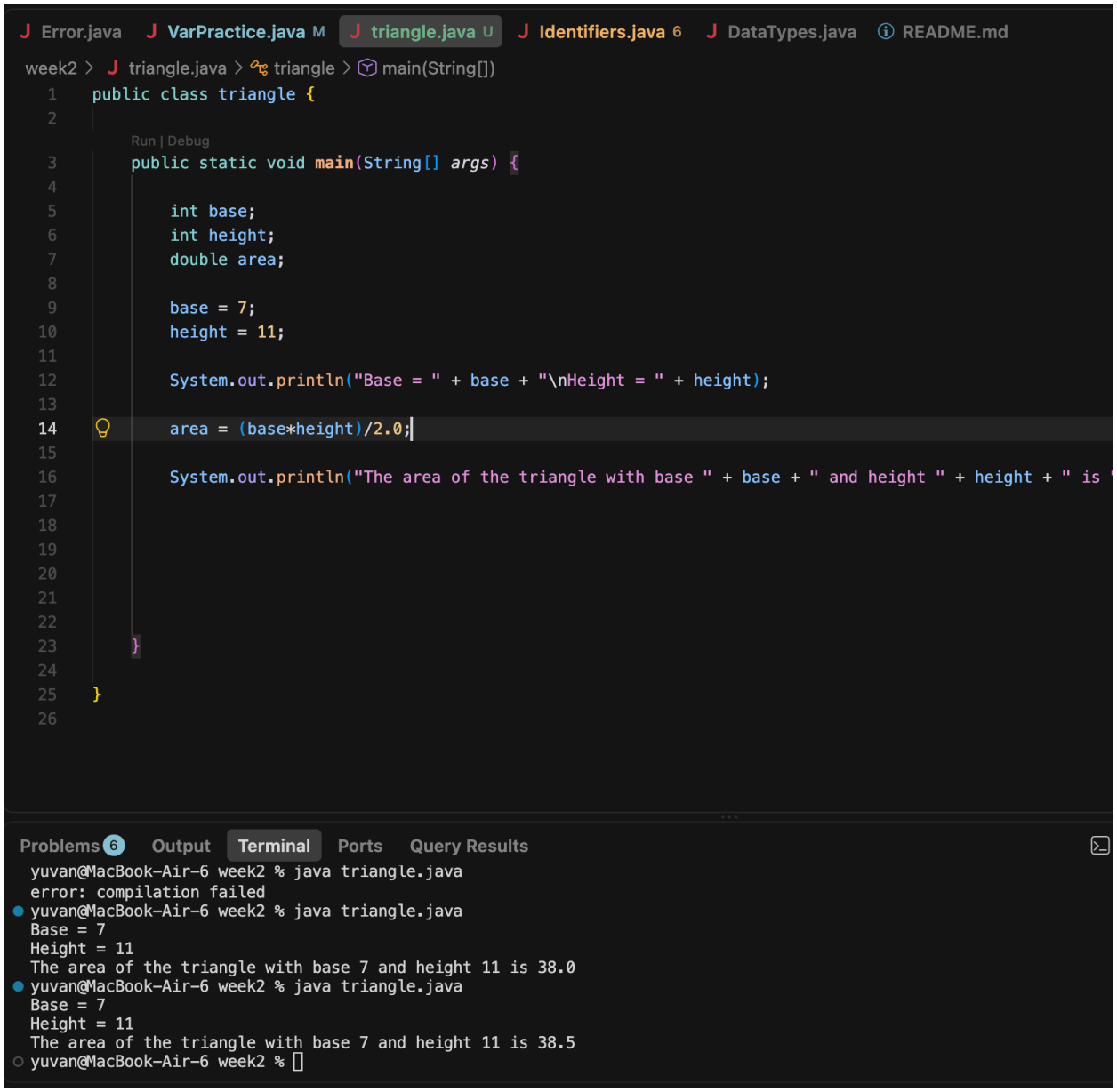
August 17 Exit Questions:

1. Syntax errors are when you don't write the code correctly, and the compiler cannot read the code that you write. A runtime error is something that breaks your code when you run the code (divide by zero).
2. If you have a syntax error, the code will not compile
3. 3. If you have a runtime error, the code will still compile
4. Semantic errors are the hardest to troubleshoot because it does not tell you where or what is breaking. The code compiles, but it does not do what you want it to do. You must find the logic error on your own within the code.

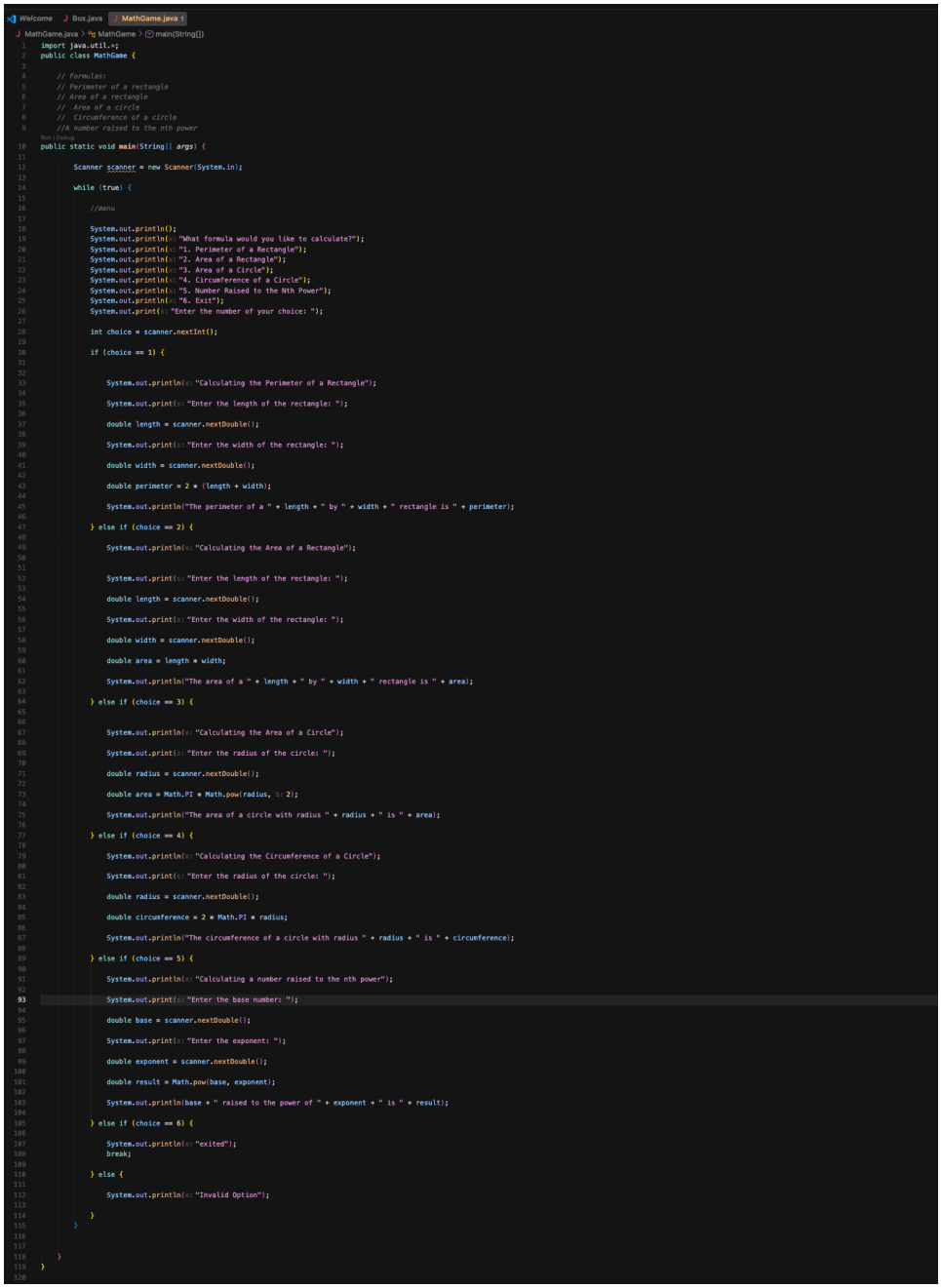
8/18/26 Classwork - VarPractice



8/20/26 Warmup



8/20/26 Classwork(Math Game) Code and Output:



```java
import java.util.*;
public class MathGame {

    // Formulas:
    // Perimeter of a rectangle
    // Area of a rectangle
    //  Area of a circle
    //  Circumference of a circle
    //A number raised to the nth power
Run | Debug
public static void main(String[] args) {

        Scanner scanner = new Scanner(System.in);

        while (true) {

            //menu

            System.out.println();
            System.out.println("What formula would you like to calculate?");
            System.out.println("1. Perimeter of a Rectangle");
            System.out.println("2. Area of a Rectangle");
            System.out.println("3. Area of a Circle");
            System.out.println("4. Circumference of a Circle");
            System.out.println("5. Number Raised to the Nth Power");
            System.out.println("6. Exit");
            System.out.print("Enter the number of your choice: ");

            int choice = scanner.nextInt();

            if (choice == 1) {


                System.out.println("Calculating the Perimeter of a Rectangle");

                System.out.print("Enter the length of the rectangle: ");

                double length = scanner.nextDouble();

                System.out.print("Enter the width of the rectangle: ");

                double width = scanner.nextDouble();

                double perimeter = 2 * (length + width);

                System.out.println("The perimeter of a " + length + " by " + width + " rectangle is " + perimeter);

            } else if (choice == 2) {

                System.out.println("Calculating the Area of a Rectangle");


                System.out.print("Enter the length of the rectangle: ");

                double length = scanner.nextDouble();

                System.out.print("Enter the width of the rectangle: ");

                double width = scanner.nextDouble();

                double area = length * width;

                System.out.println("The area of a " + length + " by " + width + " rectangle is " + area);

            } else if (choice == 3) {


                System.out.println("Calculating the Area of a Circle");

                System.out.print("Enter the radius of the circle: ");

                double radius = scanner.nextDouble();

                double area = Math.PI * Math.pow(radius, 2);

                System.out.println("The area of a circle with radius " + radius + " is " + area);

            } else if (choice == 4) {

                System.out.println("Calculating the Circumference of a Circle");

                System.out.print("Enter the radius of the circle: ");

                double radius = scanner.nextDouble();

                double circumference = 2 * Math.PI * radius;

                System.out.println("The circumference of a circle with radius " + radius + " is " + circumference);

            } else if (choice == 5) {

                System.out.println("Calculating a number raised to the nth power");

                System.out.print("Enter the base number: ");

                double base = scanner.nextDouble();

                System.out.print("Enter the exponent: ");

                double exponent = scanner.nextDouble();

                double result = Math.pow(base, exponent);

                System.out.println(base + " raised to the power of " + exponent + " is " + result);

            } else if (choice == 6) {

                System.out.println("exited");
                break;

            } else {

                System.out.println("Invalid Option");

            }
        }

    }
}
```

Output is Below for MathGame

```
yuvan@MacBook-Air-6 week2 % java MathGame
The perimeter of a 3.0 by 2.0 rectangle is 10.0

What formula would you like to calculate?
1. Perimeter of a Rectangle
2. Area of a Rectangle
3. Area of a Circle
4. Circumference of a Circle
5. Number Raised to the Nth Power
6. Exit
Enter the number of your choice: 2
Calculating the Area of a Rectangle
Enter the length of the rectangle: 1
Enter the width of the rectangle: 2
The area of a 1.0 by 2.0 rectangle is 2.0

What formula would you like to calculate?
1. Perimeter of a Rectangle
2. Area of a Rectangle
3. Area of a Circle
4. Circumference of a Circle
5. Number Raised to the Nth Power
6. Exit
Enter the number of your choice: 3
Calculating the Area of a Circle
Enter the radius of the circle: 4
The area of a circle with radius 4.0 is 50.26548245743669

What formula would you like to calculate?
1. Perimeter of a Rectangle
2. Area of a Rectangle
3. Area of a Circle
4. Circumference of a Circle
5. Number Raised to the Nth Power
6. Exit
Enter the number of your choice: 4
Calculating the Circumference of a Circle
Enter the radius of the circle: 5
The circumference of a circle with radius 5.0 is 31.41592653589793

What formula would you like to calculate?
1. Perimeter of a Rectangle
2. Area of a Rectangle
3. Area of a Circle
4. Circumference of a Circle
5. Number Raised to the Nth Power
6. Exit
Enter the number of your choice: 5
Calculating a number raised to the nth power
Enter the base number: 4
Enter the exponent: 2
4.0 raised to the power of 2.0 is 16.0

What formula would you like to calculate?
1. Perimeter of a Rectangle
2. Area of a Rectangle
3. Area of a Circle
4. Circumference of a Circle
5. Number Raised to the Nth Power
6. Exit
Enter the number of your choice: 6
exited
yuvan@MacBook-Air-6 week2 %
```